MARWAN ASHRAF

AI Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist with experience developing machine learning, deep learning, NLP, and computer vision solutions. Skilled in designing predictive models, data pipelines, and AI applications using Python, SQL, PyTorch, and Scikit-learn. **Previous leadership experience in marketing and business strategy strengthens the ability to translate business challenges into data-driven AI solutions.**

SKILLS AND ABILITIES

Programming

Python • SQL • R • PL/SQL • C++ • Java • C# • Scala

Machine Learning

Supervised & Unsupervised Learning • Regression • Classification • Clustering • Ensemble Methods
Feature Engineering • Feature Selection • Model Evaluation • Cross-Validation

Deep Learning

PyTorch • CNN • Vision Transformers (Vit) • NLP • Transformers • SBERT

Natural Language Processing

Transformers • Hugging Face • Sentence Transformers (SBERT) • Text Classification • Text Embeddings
Semantic Search • Speech-to-Text (Whisper) • Text Summarization

Computer Vision

Image Classification, Object Detection, Image Preprocessing, Data Augmentation, Feature Extraction

Data Science & Analytics

Exploratory Data Analysis (EDA) • Scikit-learn • XGBoost • CatBoost • Pandas • NumPy

Data Engineering

ETL • Data Warehousing • SQL Server • Data Pipelines • Web Scraping

Deployment & Development

Git • Streamlit • Hugging Face • BeautifulSoup • Librosa

EXPERIENCE

AI Engineer | Data Scientist | Upwork **JUN 2025 – Present**

- Built and deployed scalable end-to-end AI and Machine Learning solutions for clients using Python, Scikit-Learn, and PyTorch.
- Automated data cleaning and feature extraction pipelines to improve accuracy and efficiency.
- Build Machine Learning, Deep Learning, Computer Vision, and NLP solutions tailored to client business requirements.

Marketing Manager | Seaside Beach Resort | Zanzibar, Tanzania **JUL 2026 - Present**

- Lead digital marketing strategy, brand positioning, and customer acquisition initiatives for a luxury hospitality brand.
- Develop data-driven marketing reports and performance dashboards to measure campaign effectiveness.
- Collaborate with management teams to align marketing objectives with business goals and guest experience initiatives.

Marketing Manager | Matemwe Sands Beach Resort | Zanzibar, Tanzania **JUN 2025 – JUL 2026**

- Lead marketing team into for filling the company's marketing objectives, using technology, modern methods, not only to reach more audience, but also to embrace the brand name.
- Analyze campaign performance, customer behavior, and market trends to support revenue growth and strategic decision-making.
- Oversee content strategy, social media presence, and digital advertising activities.

Data Scientist | Intern | Support ASU **JUL 2025 – SEP 2025**

- Conducted exploratory data analysis (EDA) and applied statistical modeling techniques.
- Improved predictive accuracy of models through feature selection and hyperparameter tuning.
- Developed visualizations and analytical reports to communicate insights to technical and non-technical stakeholders.

Marketing Specialist | Al Sahel Travel Agency | Cairo, Egypt **JUN 2024 - MAR 2025**

- Directed marketing strategy, brand development, and customer engagement initiatives for a travel agency.
- Managed digital marketing campaigns across multiple online channels.
- Conducted market research, competitor analysis, and customer segmentation studies.
- Monitored campaign KPIs and optimized marketing performance using analytical insights.
- Supported business growth through lead generation and customer acquisition strategies.

Data Engineer | Intern | POTENTIA ASU **JUL 2024 – SEP 2024**

- Designed and implemented data warehouse architecture for analytical reporting.
- Developed ETL pipelines to extract, transform, and load data from multiple sources.
- Built SQL-based dashboards and reporting systems to track KPIs and business performance.

EDUCATION

Ain Shams University – Cairo, Egypt **2026**

Computer Science Degree – **Scientific Computing** Department

Relevant Coursework:

Machine Learning • Deep Learning • Computer Vision • Natural Language Processing • Statistical Analysis • Digital Signal Processing (DSP) • Human–Computer Interfaces (HCI) • Computational Robotics • Quantum Computing • Data Structures & Algorithms • Database Systems • SWE

CERTIFICATIONS

- Data Science Training – **Support ASU** (2025)
- Artificial Intelligence Fundamentals – **IBM** (2025)
- Machine Learning Specialization – **DeepLearning.AI & Stanford** (2025)
- Introduction to Statistics – **Stanford University** (2025)
- Data Science: R Programming Complete Diploma – **Udemy** (2024)
- IBM Data Analyst Career Certificate – **IBM** (2024)
- SQL for Data Analysis with SQL Server – **Udemy** (2023)
- POTENTIA Data Engineering Summer Training – **POTENTIA FCIS-ASU** (2023)
- Coding with Python – **GOMYCODE** (2022)

LANGUAGES

Arabic: Native **English**: Fluent **French**: Beginner

PROJECTS

UAE Government Customer Experience Analytics Platform

Power BI | Data Analytics | Business Intelligence

- Developed a customer experience analytics solution for a major UAE governmental institution.
- Designed KPI scoring models and performance measurement frameworks across multiple digital channels.
- Applied Machine Learning models for customer segmentation and performance forecasting to identify trends and support strategic planning.
- Built interactive Power BI dashboards and automated reports to support service quality assessment and data-driven decision-making.

Smart Retina – AI-Powered Retinal Disease Classification System

(Graduation Project – A+) Web & Mobile Applications
Deep Learning | Computer Vision | PyTorch | Streamlit

- Developed an end-to-end AI system for automated retinal disease classification using fundus images across 9 retinal conditions.
- Designed a complete pipeline including EDA, preprocessing, image standardization, model training, and deployment-ready inference.
- Implemented and evaluated EfficientNet-B0 and ResNet-50 architectures using transfer learning and GPU acceleration.
- Achieved 92.96% validation accuracy with EfficientNet-B0 and deployed a real-time diagnostic prototype using Streamlit.

Road Sign Detection & Recognition System
Computer Vision | MATLAB | Image Processing

- Developed a road sign detection and recognition system for autonomous driving applications.
- Applied image processing techniques including segmentation, color analysis, and shape detection.
- Built an interactive MATLAB GUI for image-based and real-time sign classification.

Local RAG Platform for Document Intelligence
NLP | LLMs | RAG | FastAPI | Docker

- Developed an end-to-end Retrieval-Augmented Generation (RAG) system for querying unstructured documents.
- Implemented document parsing, chunking, embeddings, and vectorbased retrieval pipelines.
- Integrated LLM-powered response generation with context-aware retrieval. Deployed the solution through a containerized FastAPI backend using Docker.

WhisprPlus – AI Transcription Tool

- Created a transcription and summarization platform using OpenAI Whisper and HuggingFace Transformers.
- Enabled speaker detection and real-time analytics in Streamlit interface.

Marketing Campaign Analysis – Data Science

- Conducted an end-to-end marketing campaign analysis using Python and Power BI.
- Cleaned and transformed campaign data, performed segmentation and performance evaluation, and built visual dashboards to highlight ROI and audience engagement insights.

Note: You can explore more of my projects on my GitHub profile and review my certifications on my LinkedIn profile.